

Module 3 Assignment

Sunday, February 8, 2026 1:22 PM

Q1

State and explain the 2 techniques computers use to handle multiple interrupts.

1. Interrupt masking (disabling interrupts temporarily):
When the CPU begins servicing an interrupt, it can temporarily disable (mask) other interrupts. This prevents additional interrupts from occurring during the current interrupt service routine (ISR), protecting shared state and ensuring the ISR completes safely. Interrupts are re-enabled after the ISR finishes (or when it reaches a safe point).
2. Interrupt priority and nesting (priority-based handling):
Interrupts are assigned priority levels. While servicing a lower-priority interrupt, the CPU can allow a higher-priority interrupt to preempt it. The CPU saves the context of the interrupted ISR, runs the higher-priority ISR, and then restores the saved context to resume the earlier ISR.

Q2

Compute the average access time (in microseconds).

Given:

- Register hit time = $20 \mu s$
- Cache access time (if not in registers) = $40 \mu s$ to bring to registers + $20 \mu s$ restart

$$T_C = 40 + 20 = 60 \mu s$$

- Main memory (if not in cache) = 30 ms fetch + $40 \mu s$ copy to cache + $20 \mu s$ restart
Convert $30 \text{ ms} = 30,000 \mu s$:

$$T_M = 30,000 + 40 + 20 = 30,060 \mu s$$

- Disk (if not in main memory) = 720 ms disk + 30 ms memory + $40 \mu s$ + $20 \mu s$
Convert $720 \text{ ms} = 720,000 \mu s$ and $30 \text{ ms} = 30,000 \mu s$:

$$T_D = 720,000 + 30,000 + 40 + 20 = 750,060 \mu s$$

Hit ratios:

- Registers hit ratio: 0.8
- Cache hit ratio: 0.5 (given a register miss)
- Main memory hit ratio: 0.2 (given a cache miss)

Probabilities:

$$\begin{aligned} P(R) &= 0.8 \\ P(C) &= (1 - 0.8)(0.5) = 0.1 \\ P(M) &= (1 - 0.8)(1 - 0.5)(0.2) = 0.02 \\ P(D) &= (1 - 0.8)(1 - 0.5)(1 - 0.2) = 0.08 \end{aligned} \quad \left. \vphantom{\begin{aligned} P(R) &= 0.8 \\ P(C) &= (1 - 0.8)(0.5) = 0.1 \\ P(M) &= (1 - 0.8)(1 - 0.5)(0.2) = 0.02 \\ P(D) &= (1 - 0.8)(1 - 0.5)(1 - 0.2) = 0.08 \end{aligned}} \right\} 1$$

Average access time:

$$\begin{aligned} T_{avg} &= P(R) T_R + P(C) T_C + P(M) T_m + P(D) T_D \\ &= 0.8(20) + 0.1(60) + 0.02(3060) + 0.8(750060) \\ &= 16 + 6 + 601.2 + 60004.8 \end{aligned}$$

$$T_{avg} = 60628 \mu s$$

Q3

There are 83 tape drives. Each job must be allocated 5 drives immediately, and they are held until job completion. A job cannot start unless 5 drives are free. During execution, a job uses only 4 drives for a long time, then 5 drives near completion.

A) Maximum number of jobs in progress

Each job requires 5 drives reserved:

$$\left\lfloor \frac{83}{5} \right\rfloor = \lfloor 16.6 \rfloor = \underline{16}$$

B) Maximum and minimum number of tape drives idle due to this policy

If 16 jobs are admitted:

- Drives allocated = $16 \cdot 5 = 80$
- Drives unallocated (cannot start another job) = $83 - 80 = 3$

Maximum idle: when each of the 16 jobs is in the long phase using only 4 drives:

- Idle-but-held per job = 1
- Total idle-but-held = $16 \cdot 1 = 16$
- Plus the 3 unallocated drives:

$$16 + 3 = \underline{19}$$

Minimum idle: when all 16 jobs are in the phase using all 5 drives:

- No idle-but-held drives
- Only the 3 unallocated drives remain idle

Q4

Cache access time $T_c = 50$ ns, main memory access time $T_m = 1000$ ns. Effective access time (EAT) is 30% more than cache.

$$EAT = 1.3 \times 50 = 65 \text{ ns}$$

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$$EAT = hT_c + (1-h)(T_c + T_m)$$

$$\Rightarrow 65 = h(50) + (1-h)(50 + 1000)$$

$$\Rightarrow 65 = 50h + 1050 - 1050h$$

$$\Rightarrow 65 = 1050 - 1000h$$

$$\Rightarrow 1000h = 985$$

$$\Rightarrow h = 0.985$$